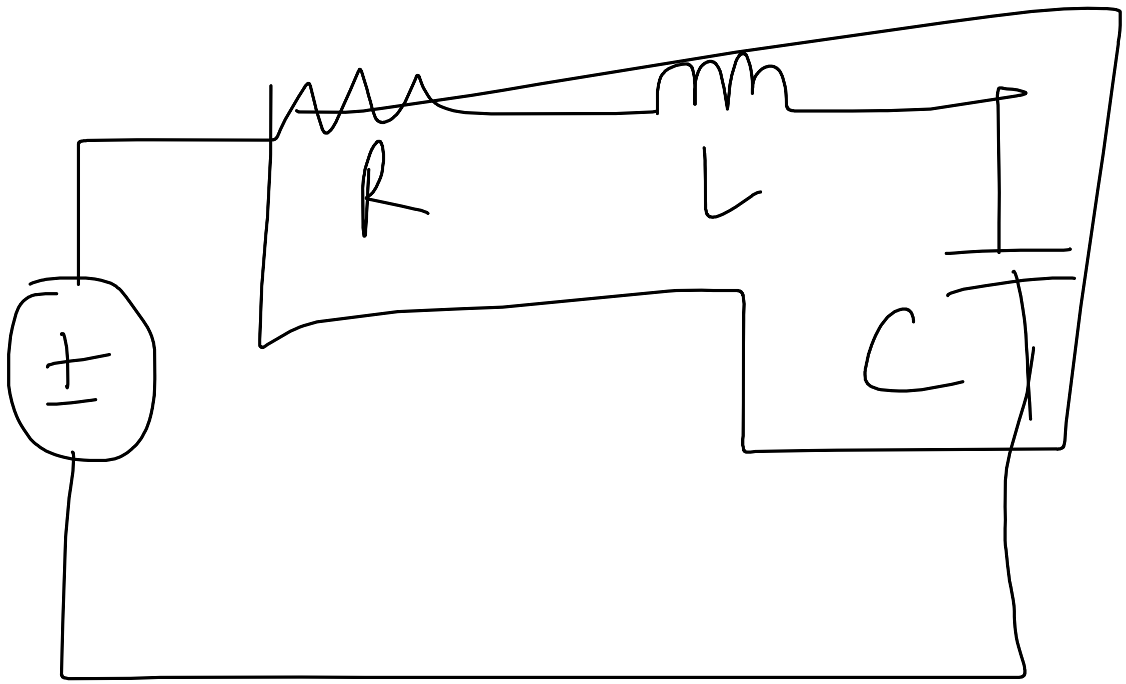
Series Resonance



Series RLC cnt

$L \rightarrow Z_L$

$C \rightarrow Z_C$

$$\boxed{Z = R + j\left(\omega L - \frac{1}{\omega C}\right)}$$

Special cond^n $\longrightarrow$ $\boxed{\text{resonance}}$

$$\boxed{\mathrm{Im}(Z) = 0}$$

$$\omega L - \frac{1}{\omega C} = 0$$

$$\Rightarrow \omega^2 = \frac{1}{LC}$$

$$\therefore \boxed{\omega_0 = \frac{1}{\sqrt{LC}}}$$

$Z = R + j(\;\;)^0$

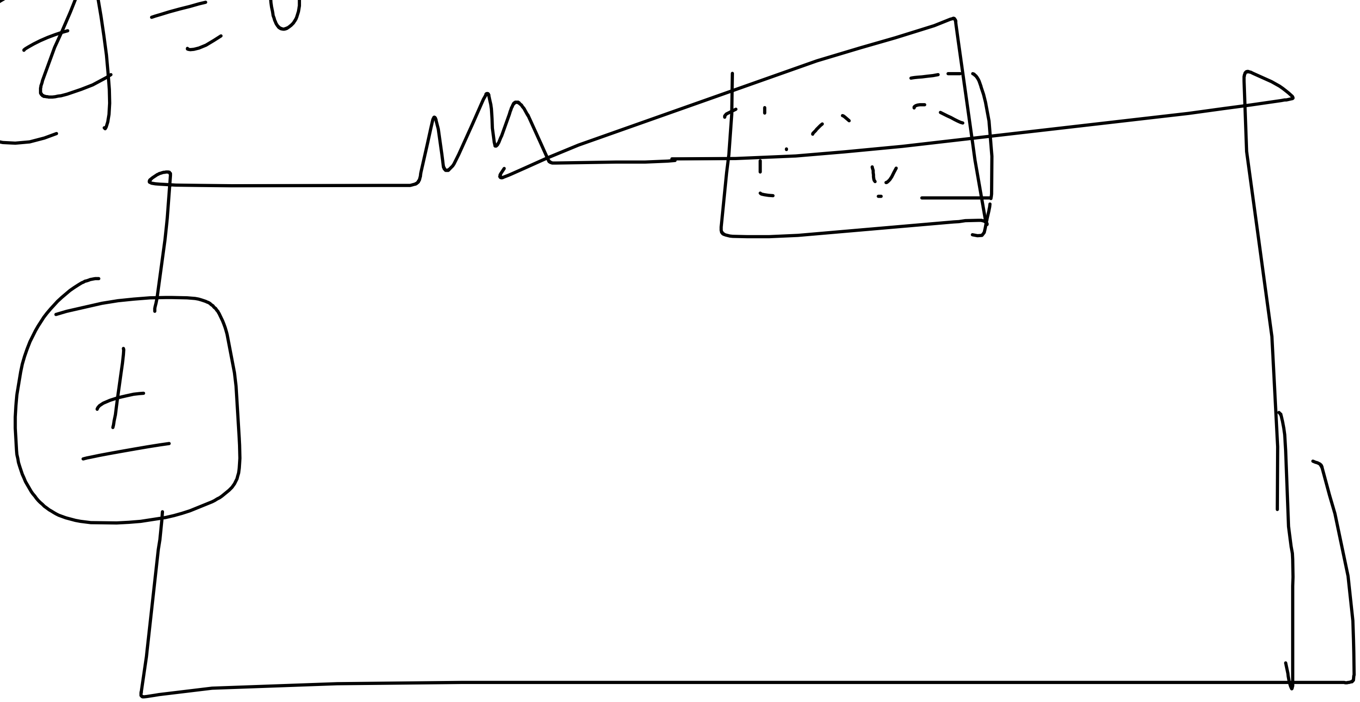
$Z = R$

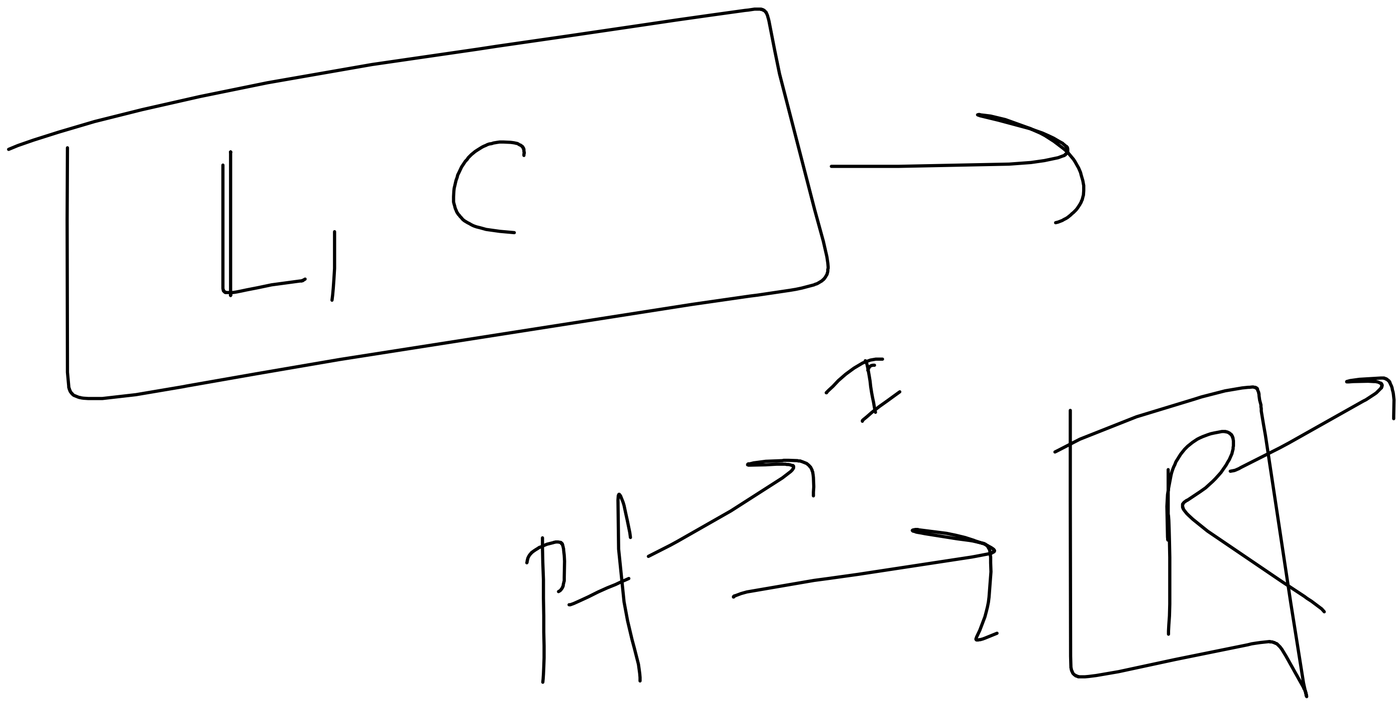
more energy efficient

$V = \frac{1}{R}$

At resonance cond^n

$Im[Z] = 0$




L, C
H
I
R

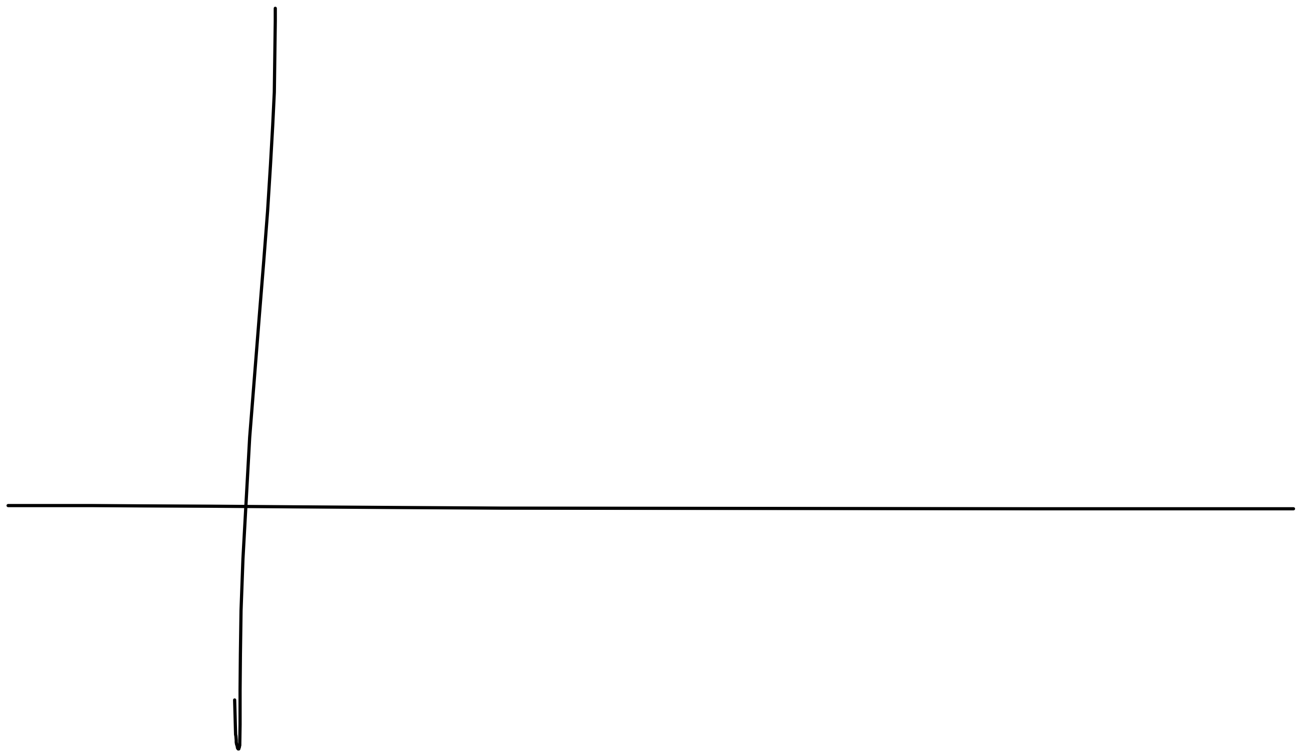
① Energy Efficiency

$P f \rightarrow 1$ unitity

$\frac{1}{\boxed{P}} \approx 1$

$\boxed{Q = 0}$

$\boxed{Z = R}$ ②



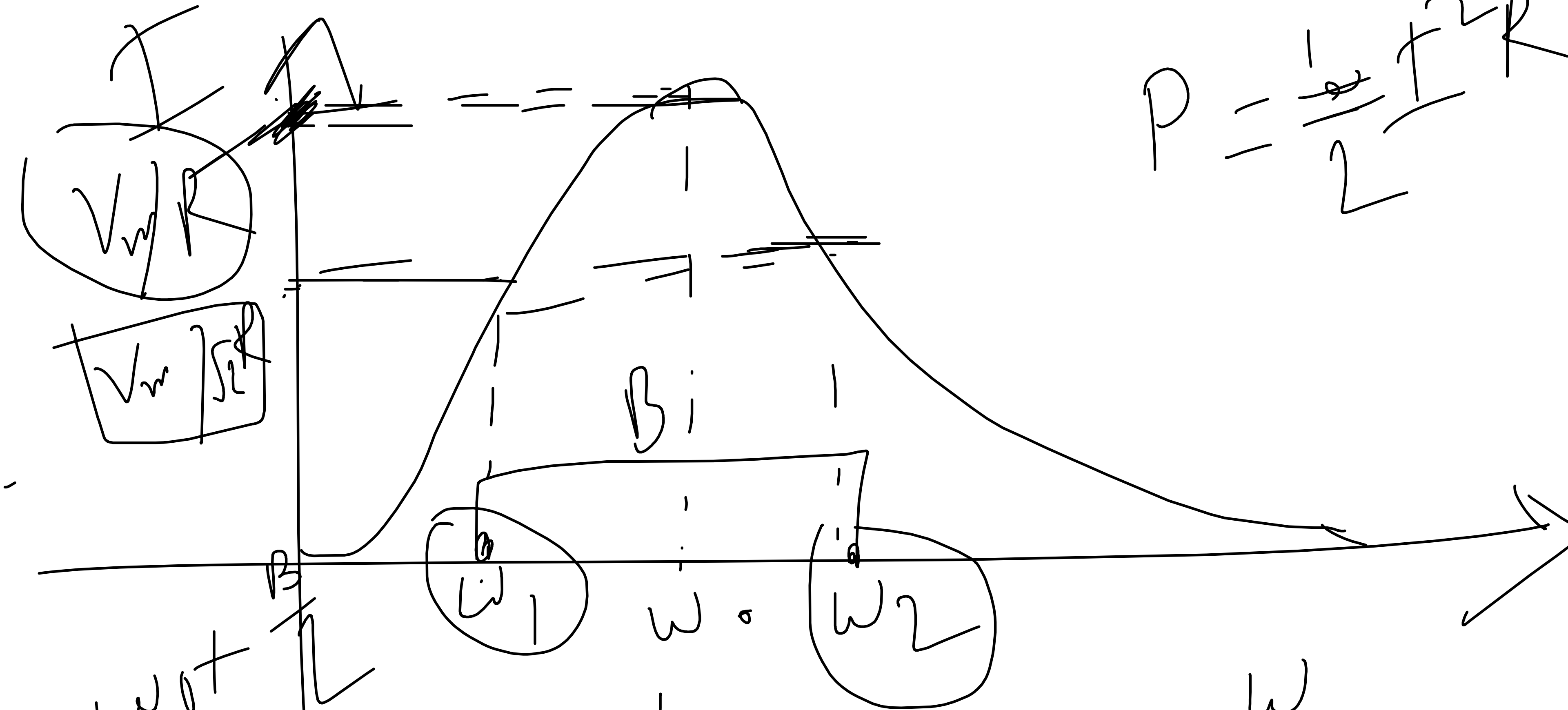
$$Z = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$$

constant

$$V = IR$$

$$|I| = \frac{|V|}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}}$$

$P = \frac{1}{2} I^2 R$



$\omega_2 = \omega_0 + \frac{B}{2}$

$\frac{1}{\sqrt{LC}}$

Solving for ω, we obtain

$$\boxed{\begin{aligned}\omega_1 &= -\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}} \\ \omega_2 &= \frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}\end{aligned}} \qquad \text{(14}$$

half frequancy

$\omega_0 \rightarrow$ Central freq

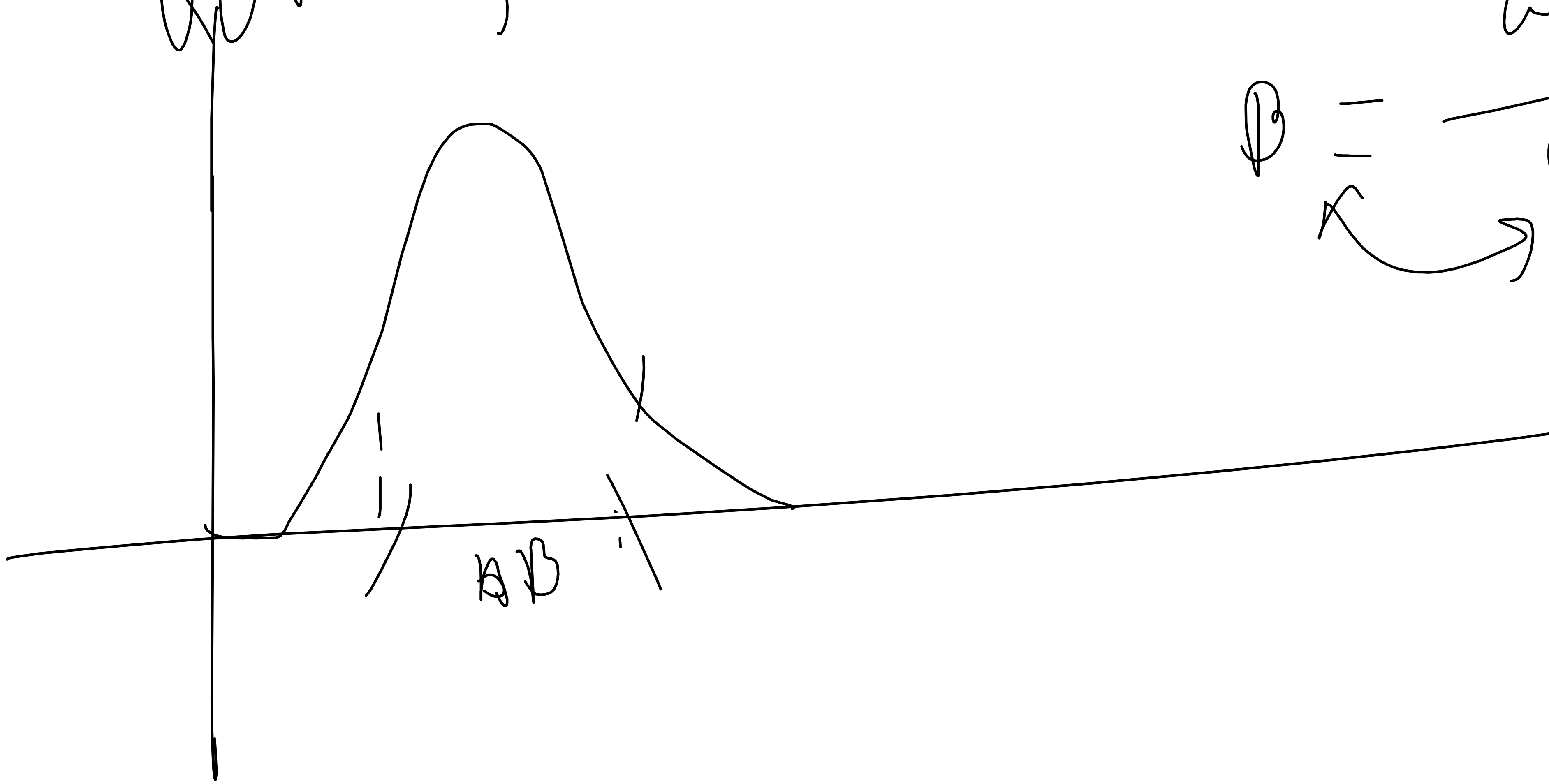
Resonant n

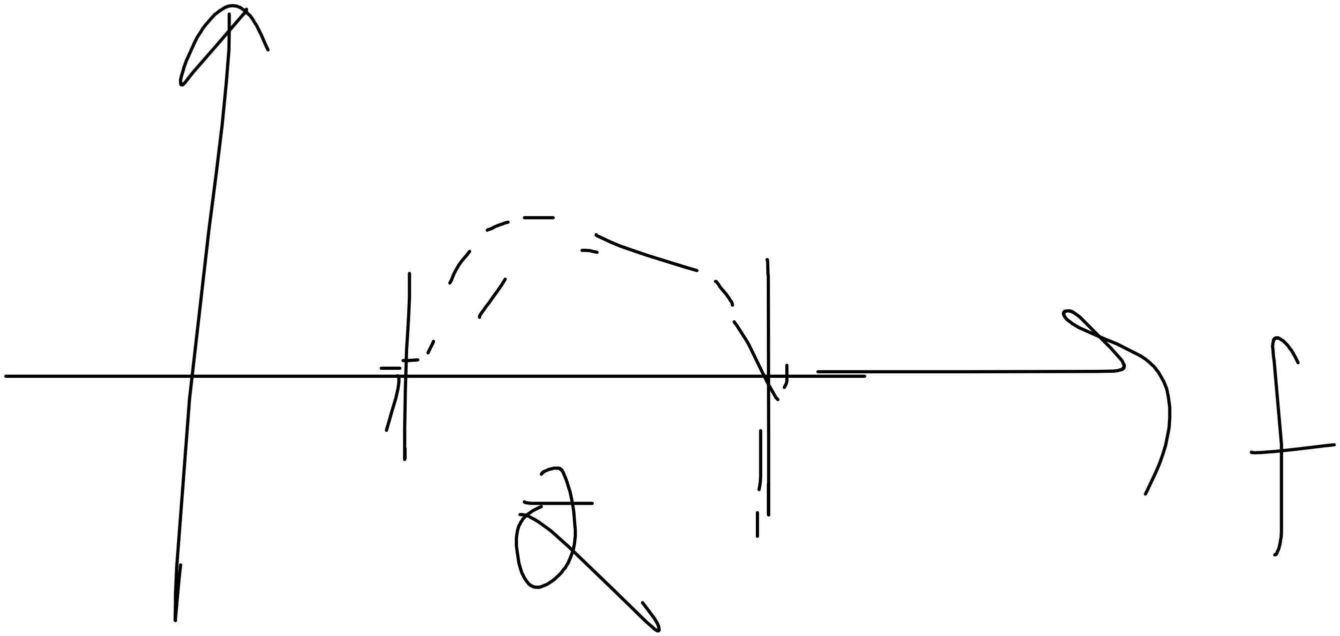
$$\boxed{\omega_0 = \sqrt{\omega_1 \omega_2}}$$

$$\underline{\omega_2 - \omega_1 = B}$$

Quality factor $\rightarrow Q$

$Q = \frac{\omega_0}{\;}$

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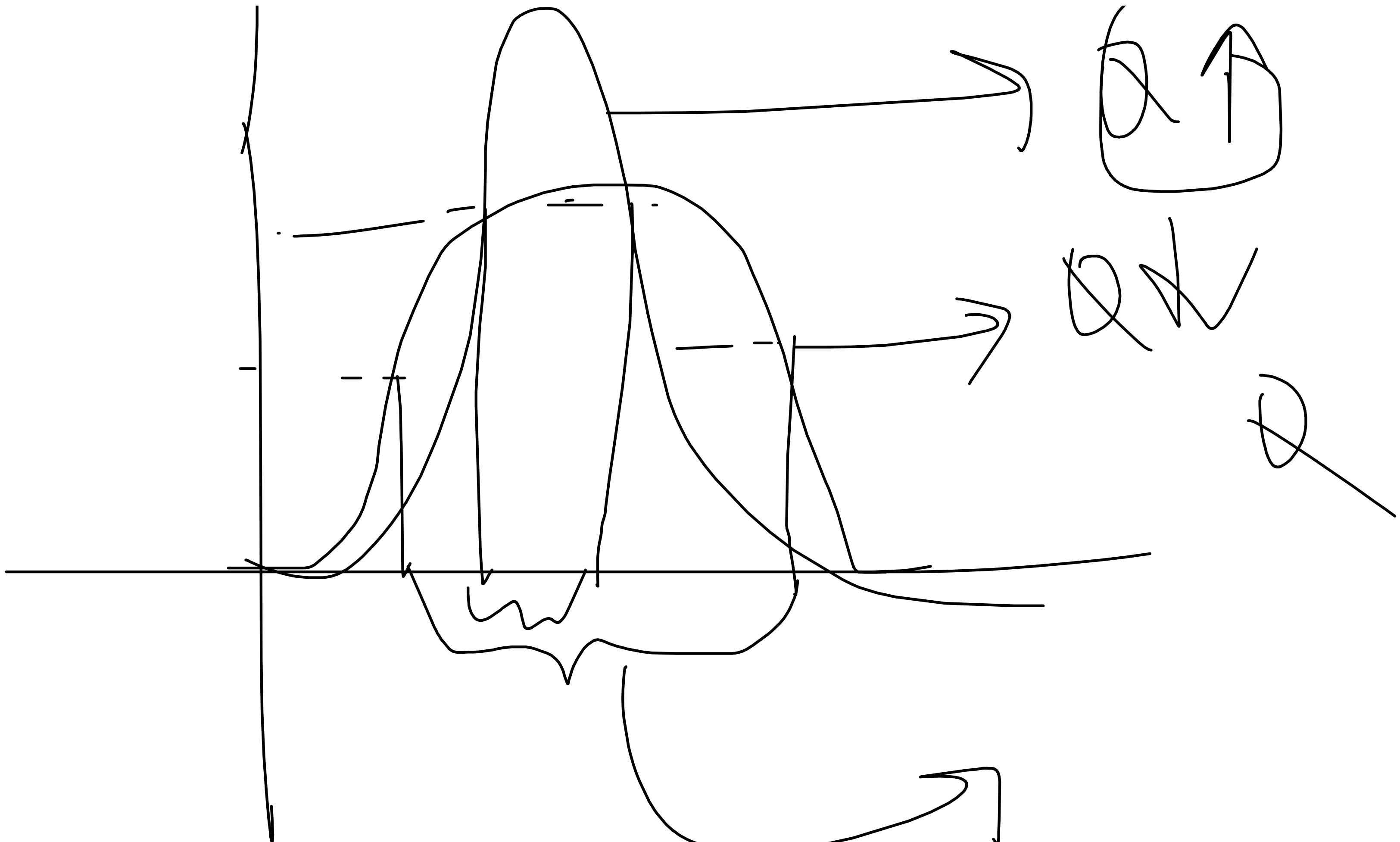


$$Q = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 C R}$$

$$B = \frac{\omega_0}{Q}$$

Practice Problem 14.7

A series-connected circuit has $R = 4\ \Omega$ and $L = 25$ mH. (a) Calculate the value of C that will produce a quality factor of 50. (b) Find ω_1, ω_2, and B. (c) Determine the average power dissipated at $\omega = \omega_0, \omega_1, \omega_2$. Take $V_m = 100$ V.

$$Q = \frac{\omega_0 L}{R} = \boxed{\frac{1}{\omega_0 CR}} \qquad (1$$

at the quality factor is dimensionless. The relatic
he bandwidth B and the quality factor Q is obtained b
q. (14.33) into Eq. (14.35) and utilizing Eq. (14.38).

$$B = \frac{R}{L} = \frac{\omega_0}{Q} \qquad (1$$

$$\boxed{\omega_1 \simeq \omega_0 - \frac{B}{2}, \qquad \omega_2 \simeq \omega_0 + \frac{B}{2}} \qquad (14$$

circuits are used often in communications networks.

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

Here,

$$Q = 50$$

$$Q = \frac{1}{\boxed{\omega_0} CR}$$

$$Q = \frac{w_0 L \phi}{R}$$

$$\therefore w_0 = \frac{QR}{L}$$

$$= \frac{50 \times 4}{25 \times 10^{-3}}$$

$$= 8 \times 10^{3} \text{ rad/s}$$

$$Q = \frac{1}{\omega_0 C R}$$

$$\therefore C = \frac{1}{Q \omega_0 R}$$

$$= 0.625 \mu F.$$

(ii)

$$B = \frac{R}{L}$$

$$= \frac{4}{25m}$$

$$= 160 \text{ rad/s}$$

$$w_1 = w_0 - \frac{B}{2}$$

$$= 6\times10^3 - 80$$

$$w_2 = w_0 + \frac{B}{2}$$

$$= 6\times10^3 + 80$$

$$=$$

(iii) $\omega = \omega_0$ $P = \dfrac{V^2}{2R}$

$\dfrac{V^2}{R}$

$= 1250\ W.$

$\omega = \omega_1, \omega_2$ in half freq

at half freq. power is halved

$\therefore P = \dfrac{1}{2} \times 1250$

$= 625\ W.$

Parallel Renonance

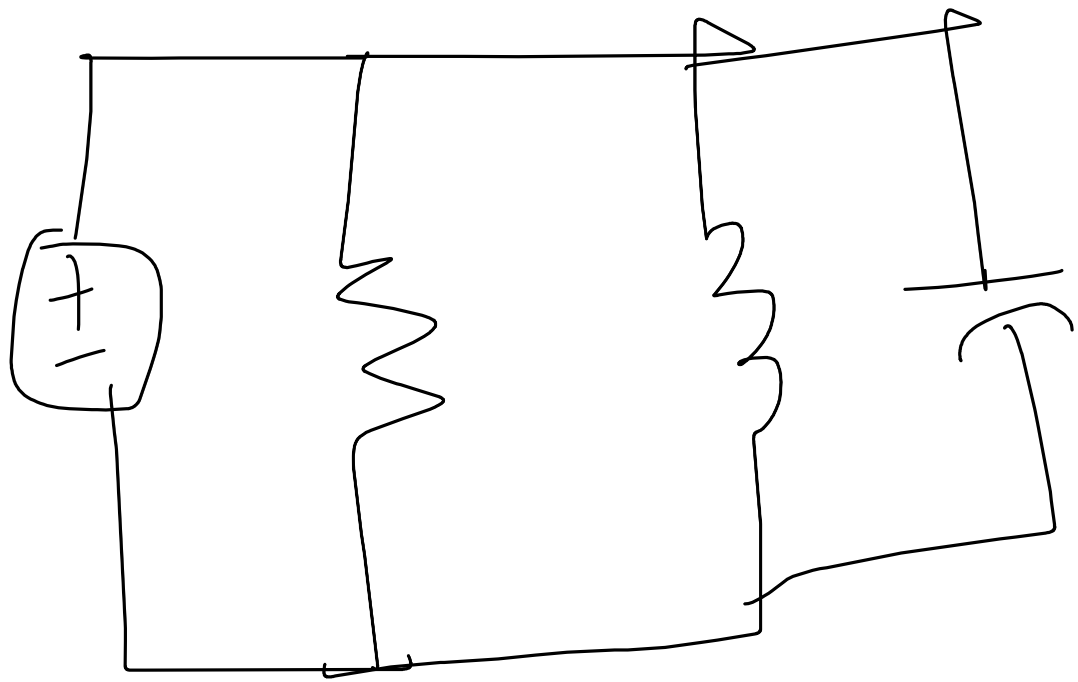


TABLE 14.4

Summary of the characteristics of resonant *RLC* circuits.

Characteristic	Series circuit	Parallel circuit
Resonant frequency, ω_0	$\dfrac{1}{\sqrt{LC}}$	$\dfrac{1}{\sqrt{LC}}$
Quality factor, Q	$\dfrac{\omega_0 L}{R}$ or $\dfrac{1}{\omega_0 RC}$	$\dfrac{R}{\omega_0 L}$ or $\omega_0 RC$
Bandwidth, B	$\dfrac{\omega_0}{Q}$	$\dfrac{\omega_0}{Q}$
Half-power frequencies, ω_1, ω_2	$\omega_0\sqrt{1+\left(\dfrac{1}{2Q}\right)^2} \pm \dfrac{\omega_0}{2Q}$	$\omega_0\sqrt{1+\left(\dfrac{1}{2Q}\right)^2} \pm \dfrac{\omega_0}{2Q}$
For $Q \geq 10$, ω_1, ω_2	$\omega_0 \pm \dfrac{B}{2}$	$\omega_0 \pm \dfrac{B}{2}$

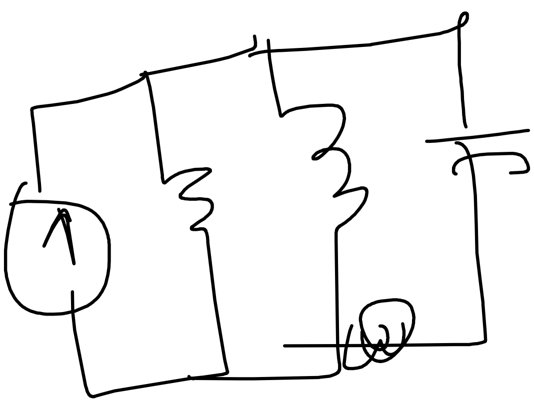
Practice Problem 14.8 A parallel resonant circuit has $R = 100\ \text{k}\Omega$, $L = 20$ mH, and $C = 5$ nF. Calculate ω_0, ω_1, ω_2, Q, and B.

TABLE 14.4

Summary of the characteristics of resonant RLC circuits.

Characteristic	Series circuit	Parallel circuit
Resonant frequency, ω_0	$\frac{1}{\sqrt{LC}}$	$\frac{1}{\sqrt{LC}}$
Quality factor, Q	$\frac{\omega_0 L}{R}$ or $\frac{1}{\omega_0 RC}$	$\frac{R}{\omega_0 L}$ or $\omega_0 RC$
Bandwidth, B	$\frac{\omega_0}{Q}$	$\frac{\omega_0}{Q}$
Half-power frequencies, ω_1, ω_2	$\omega_0\sqrt{1+\left(\frac{1}{2Q}\right)^2} \pm \frac{\omega_0}{2Q}$	$\omega_0\sqrt{1+\left(\frac{1}{2Q}\right)^2} \pm \frac{\omega_0}{2Q}$
For $Q \geq 10$, ω_1, ω_2	$\omega_0 \pm \frac{B}{2}$	$\omega_0 \pm \frac{B}{2}$

$$\omega_0 = \frac{1}{\sqrt{LC}} = 1 \times 10^5 \text{ rad/s}$$



$$Q = \omega_0 RC = 50 \boxed{> 10}$$

$$B = \frac{\omega_0}{Q} = 2000 \text{ rad/s}$$

$$w_1 = w_0 - \frac{B}{2}$$

$$= 1 \times 10^5 - 1000$$

$$w_2 = w_0 + \frac{B}{2}$$

$$= 1 \times 10^5 + 1000$$

Example 14.9

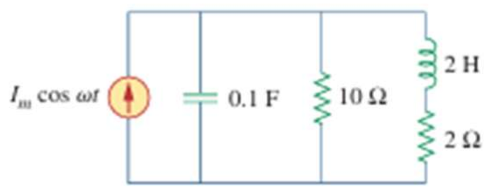


Figure 14.28
For Example 14.9.

$$\boxed{w_0 = \frac{1}{\sqrt{LC}}}$$

Real + Im.

Z Find Z =

② $\boxed{Im(Z) = 0}$

$w_0 =$